

Ngô Minh Chung

Smart Contract Engineer

Security-first Web3 Engineer focused on production-grade Solidity development, protocol auditing, and AI-driven development workflows.



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SKILLS & LANGUAGES

Blockchain Core	Solidity, EVM Architecture, Yul, ERC-20 / 721 / 4626.	Languages (Dev)	Solidity, TypeScript, JavaScript, Rust, Python.
Frameworks & Tools	Foundry, Hardhat, Ethers.js, Slither, Mythril.	Chinese	Conversational – Ready to collaborate with global Chinese Web3 dev teams.
AI-Assisted Dev	AI Agent Engineering, Autonomous Agent Loops, OpenAI/Google AI tooling.	English	Professional Reading & Technical Research.

WORK EXPERIENCE

Smart Contract & Security Intern / Apps Cyclone

Blockchain Solution Development Company

02/2026 – Present

Ho Chi Minh City, Vietnam

- **Secure Engineering:** Developed and refactored secure Solidity contracts (ERC-20/721) with zero post-deployment vulnerabilities.
- **Advanced Testing:** Built unit/fuzzing test suites using Foundry and Hardhat, achieving **>95%** test coverage.
- **Gas Optimization:** Optimized storage layouts and utilized custom errors, cutting execution gas fees by up to **15%**.

FEATURED PROJECTS

DeFi-Vault System

NFT-Powered Term Deposit Protocol

2026

Solidity, Foundry, TS, ERC-721

- **Architecture:** Separated business logic (*SavingCore*) from storage (*VaultManager*) to fully isolate execution risks.
- **Precision Math:** Handled time-based yield distribution via Basis Points (BPS) to prevent rounding and division bugs.
- **Fuzz Validation:** Applied Foundry invariant and property-based fuzzing to guarantee math correctness under edge cases.

ZK Panagram Game

Privacy-Preserving Word Game Protocol

2026

Noir, Solidity, Foundry, UltraPlonk, ERC-1155

- **ZK-Circuit Design:** Engineered a Noir circuit using *pedersen_hash* constraints to verify secret answers without exposing plain text.
- **Verifier Integration:** Hand-crafted an *UltraVerifier* contract matching standard bytes32 public inputs to bypass tool deprecation bugs.
- **On-Chain Security:** Combined game state management with anti-front-running logic and automated ERC-1155 multi-token reward minting.

Sentinel X (Build X Hackathon)

AI-Powered Real-Time Threat Detection Engine

2026

Node.js, Ethers.js, X Layer, AI Agent

- **Mempool Scanner:** Developed a runtime transaction scanner on L2 X Layer to detect honeypot and front-running signatures.
- **AI Agent Workflow:** Integrated autonomous agent loops to evaluate protocol transaction payloads against static risk matrices.
- **Auto Defense:** Engineered an automated loop using Ethers.js to trigger instant capital flight to USDT via Uniswap routers.

SECURITY RESEARCH & ACHIEVEMENTS

Web3 Security Competitor & CTF Researcher

2024 – 2026

Platforms: Ethernaut (OpenZeppelin), QnQSec, Securinets

Independent Researcher

- **Securinets Friendly CTF:** Achieved **Rank 24th Globally** (Rank 13th Friendly), solving 78 challenges for 23,030 total points.
- **QnQSec CTF 2025:** Secured **260th place** globally against highly competitive international security teams.
- **Ethernaut Wargame:** Successfully compromised over 20+ simulation levels, gaining mastery over advanced vulnerabilities (Reentrancy, Integer Overflow/Underflow, Delegatecall context-switching, and Oracle Manipulation).

Professional Training & Certifications

2025 – 2026

Cyfrin Updraft & Google AI Ecosystem

Credentials Earned

- **Google AI Certified Student:** Credentialed in fundamental AI competencies and advanced prompt engineering concepts for developer efficiency.
- **Cyfrin Updraft Certificates:** Graduated from *Smart Contract Development* and *Wallet Security* curricula with honors.

CORE COMPETENCIES

Smart Contract Architecture

AI Agent Integration

Fuzz & Invariant Testing

Adversarial Audit Mindset